Homework 3 (20pts)

Due November 4 at 11:59 PM Eastern Time

Why are we doing this? The homework assignments are a low-stakes way to practice the concepts we've talked about in class and apply them to real-world problems. For this particular assignment, I would like you to (1) practice using generative network models to compare empirical network structure against and (2) implement different notions of centrality.

How will you be assessed?

What will you submit? A PDF of the homework assignment and all of the code (Jupyter notebooks or scripts) used to generate the results and figures to the Canvas assignment.

Format requirements

- The title of the homework assignment
- Your name
- The collaborators with whom you worked
- Each problem should have a separate heading
- Plots should have descriptive axis labels
- OPTIONAL helpful tips for plots: (1) it's nice when legends and axis labels follow the format "Descriptive text, symbol" (For example, you would label an axis something like, "Contagion rate, β" by writing the following in Python: `plt.xlabel(r''Contagion rate, $\beta$'')`), (2) when saving figures (with `savefig`), either save in vector graphic format (SVG, PDF) or use the `dpi` option and set it to 300 or more (I like 1,000 DPI), (3) use sufficiently large fonts, (4) a legend when multiple things are being plotted on the same plot, (5) use the `despine` command in Seaborn to remove the box around plots, and (6) the `tight_layout` command can automatically adjust the plot spacing to make enough space around the labels, etc.

The assignment

Extra credit (1 point): Use the LaTeX template (See Canvas) to write up the homework.

Question 1 (10 points): The configuration model is a useful tool for matching degree heterogeneity while randomizing all other structure.

I've attached code snippets in a Jupyter notebook that will be helpful in completing some of the assignment.

a) Read up on the network configuration model in Coscia Chapter 18.1.

b) (4 points) Code up an implementation of the network configuration model by doing the following

   i) Write a function called `configuration_model` which accepts a list (or array) of degrees as input and returns an edge list.

   ii) Check that the sum of the degrees is even and expand the list of degrees into a list of nodes, where the number of times that a node appears in this list is its degree.

iii) Form an edge list by randomly permuting the list of nodes with (`numpy.random.shuffle`) and then pairing subsequent nodes (index 0 with index 1, index 2 with index 3, etc.).

iv) Add an additional argument called `return_graph` to `configuration_model` and use this argument to allow the function to return either an edge list or a network as a NetworkX MultiGraph object.

c) (3 points) Generate a degree distribution from the truncated power law distribution (see attached code snippet) with `N=1000`, `minval=1`, `maxval=N-1`, and `alpha=-2.5`. If the sum of degrees is odd, add one to a random node. Input this degree sequence into your function and output an edge list. Are there any self-loops or multi-edges in the resulting edge list? If so, roughly what fraction of the total number of edges are multi-edges and self-loops, respectively? Average over 1,000 realizations of the configuration model. How does this compare to the `budapest_connectome` dataset (on Canvas)?

d) (3 points) Import the edge list of the `budapest_connectome` dataset (on Canvas). (i) Create a NetworkX MultiGraph object from this edge list, (ii) generate 1,000 realizations of the configuration model fitted to this edgelist, and (iii) plot a vertical line indicating the degree assortativity (this is implemented in NetworkX) of the empirical dataset and a histogram of degree assortativity values computed from the ensemble of configuration model networks. Make sure that the axes are labeled! Is the degree assortativity more or less than you'd expect at random?

Question 2 (10 points): The Medici family was a powerful political dynasty and banking family in 15th century Florence. The classic network-based explanation of their power, offered by Padgett and Ansell in 1993, claims that they established themselves as the most central players within the network of prominent Florentine families (shown in Fig. 1).

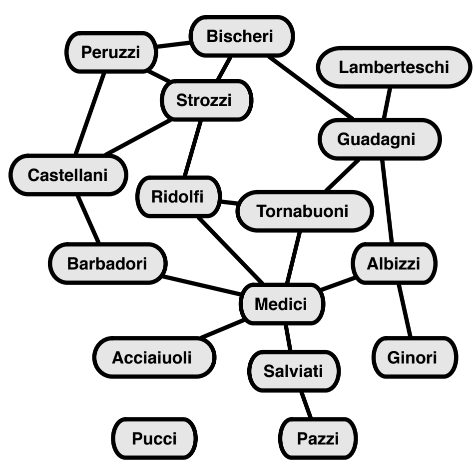

Figure 1: The Medici family alliance network, from Padgett and Ansell (1993).

Conduct the following two tests of their structural importance hypothesis:

a) (10 points; 2 point for each centrality calculation and 2 points for the discussion) Construct a table with four columns, each containing a list of (family, score) pairs for the following centrality score functions:

- degree centrality,

- harmonic centrality,
- eigenvector centrality,
- betweenness centrality,

all of which are implemented in NetworkX. Within each centrality, sort the pairs in decreasing order of importance; three decimal places is sufficient detail. Discuss (i) the degree to which these scores agree with Padgett and Ansell's claim that the Medicis occupied a structurally important position in this network, and (ii) what the scores say about the second most important family.

b) (Extra credit: 5 points) Determine whether the relative structural importance of the Medicis can be explained solely by the degree sequence of the network G. To do this, use the configuration model you implemented prior to produce an ensemble of 1,000 random graphs with the same degree sequence $\mathbf{d}$ as G, and extract a distribution of betweenness centrality scores for each node.

Make a figure showing the difference between a node's betweenness centrality on G and its average betweenness centrality in the ensemble. On this figure, include lines showing the 25% and 75% quantiles of the distribution around the mean for all vertices (as in the figure below, for the Karate club).

Finally, (i) discuss your results here in terms of Padgett and Ansell's story of the Medici family (see article on Canvas), and (ii) comment on whether your results from part a) have changed.